

MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

Bluestock Fintech Capstone Project

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Technologies Used:

Python | SQL | SQLite | Power BI | GitHub

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The Mutual Fund Analytics Platform is an end-to-end data analytics project developed as part of the Bluestock Fintech Capstone Project. The primary objective of this project is to analyze mutual fund performance, investor behavior, and industry trends using data-driven techniques.

The project involves data collection from multiple financial sources, data cleaning and transformation through an ETL pipeline, storage in a structured database, and visualization through an interactive Power BI dashboard. Various analytical techniques were applied to evaluate fund performance, monitor SIP trends, and generate meaningful insights for investment analysis.

The developed dashboard provides a centralized platform for tracking key performance indicators, comparing fund performance, and understanding investor participation trends. The project demonstrates the practical application of data analytics, database management, and business intelligence tools in the financial domain.

Project Overview

The Mutual Fund Analytics Platform is a comprehensive data analytics solution designed to analyze mutual fund performance, investor behavior, and industry trends. The project integrates data from multiple financial sources, processes it through an ETL pipeline, stores it in a structured database, and presents actionable insights through interactive Power BI dashboards.

The platform enables efficient tracking of fund performance, SIP growth, investor participation, and market trends. By combining data engineering, analytics, and visualization techniques, the project supports informed decision-making and performance evaluation within the mutual fund industry.

Project Objectives

- Develop an automated ETL pipeline for mutual fund data collection and processing.
- Design and implement a structured database for analytics.
- Perform exploratory data analysis to identify trends and patterns.
- Analyze mutual fund performance using key financial metrics.
- Study investor behavior and transaction trends.
- Create interactive dashboards for business intelligence reporting.
- Generate actionable insights to support investment analysis and decision-making.

Data Sources

The project utilizes publicly available mutual fund and financial market datasets collected from multiple sources. These datasets provide information related to fund performance, NAV history, investor transactions, SIP inflows, and benchmark indices.

The primary data sources include:

- AMFI India
- mfapi.in
- NSE India
- BSE India

The collected datasets were processed and transformed to support analytics and dashboard development.

Technology Stack

The project was developed using the following technologies:

Category	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Database	SQLite
Analytics	Jupyter Notebook
Data Visualization	Power BI
Version Control	Git & GitHub

These technologies were used to perform data extraction, transformation, analysis, storage, and visualization throughout the project lifecycle.

ETL Process

The Mutual Fund Analytics Platform follows an Extract, Transform, and Load (ETL) process to collect, process, and store financial data for analysis and reporting.

Extract

Data was collected from multiple financial data sources, including mutual fund and market-related datasets. The extraction process involved gathering data related to fund performance, NAV history, SIP inflows, investor transactions, and benchmark indices.

Transform

The extracted data underwent cleaning and transformation procedures to ensure consistency and accuracy. Missing values were handled, duplicate records were removed, and data types were standardized. Additional calculations and aggregations were performed to prepare the datasets for analysis.

Load

The transformed datasets were loaded into a structured SQLite database. The database served as the central repository for storing and managing analytics-ready data used in dashboard development and reporting.

System Architecture

The system architecture follows a structured data pipeline where raw data is collected, processed, stored, analyzed, and visualized through interactive dashboards.

Data Sources → Data Extraction → Data Cleaning & Transformation → SQLite Database → Analytics Layer → Power BI Dashboard

Dashboard Overview

An interactive Power BI dashboard was developed to provide comprehensive insights into mutual fund performance, investor behavior, and industry trends. The dashboard enables users to analyze key performance indicators, compare funds, and explore investment patterns through dynamic visualizations and filters.

The dashboard consists of four major analytical pages:

1. Industry Overview

Provides a high-level summary of the mutual fund industry, including Assets Under Management (AUM), SIP inflows, folio growth, and overall market performance.

2. Fund Performance Analysis

Displays fund returns, risk metrics, benchmark comparisons, and performance rankings to evaluate investment opportunities.

3. Investor Analytics

Analyzes investor participation, transaction behavior, and investment trends to understand investor engagement.

4. SIP & Market Trends

Presents SIP growth trends, category-wise inflows, and market movement analysis through interactive visualizations.

Dashboard – Mutual Fund AUM Performance

This dashboard highlights mutual fund industry performance through **Assets Under Management (AUM)** trends, fund house rankings, scheme strength, and market share distribution.

Key Visuals

- **AUM Trend (2023–2025):** Shows consistent industry growth.
- **Top Fund Houses:** SBI, ICICI Prudential, and HDFC lead in total AUM.
- **Scheme Strength:** Larger AMC's manage more schemes, reflecting diversification.
- **Market Share:** SBI holds the largest share (21.67%), followed by ICICI and HDFC.

Insights

- SBI dominates with the highest market share.
- ICICI and HDFC together account for over 30% of AUM.
- Smaller players like Axis and Mirae Asset show steady growth.
- Overall AUM expansion reflects strong investor participation.

Mutual Fund AUM Performance Dashboard



Dashboard – SIP Analysis

This dashboard provides insights into **Systematic Investment Plans (SIPs)**, highlighting inflows, account activity, growth rates, and assets under management. It helps track investor participation and the overall growth of SIPs in the mutual fund industry.

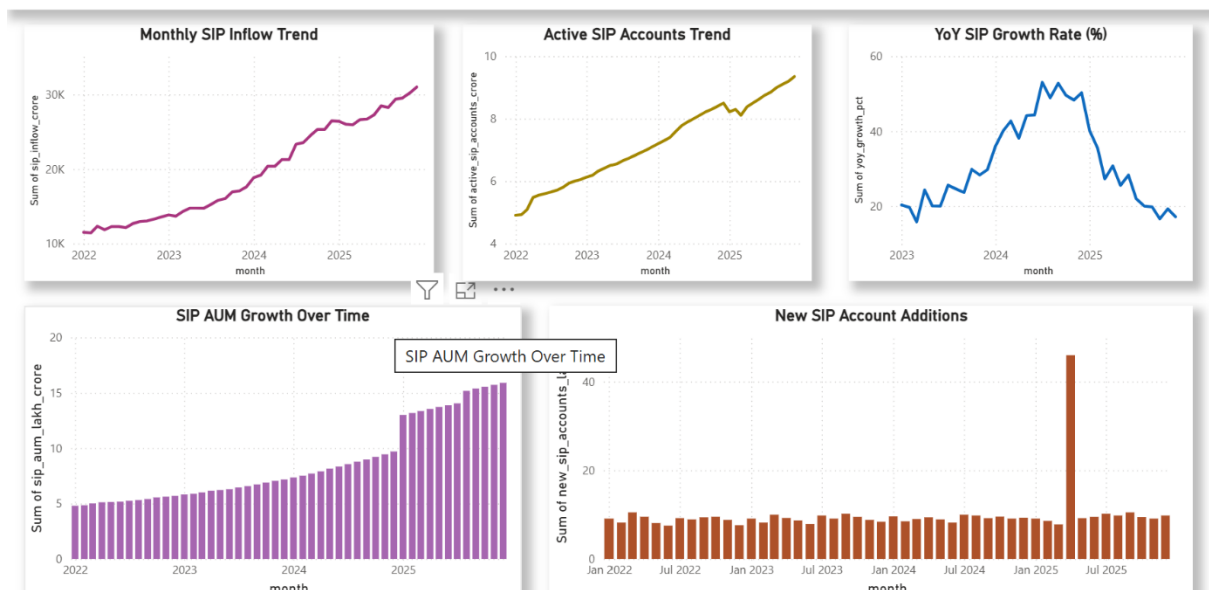
Key Visuals

- **Monthly SIP Inflow Trend:** Steady rise from 2022 to 2025, crossing ₹31,000 crore.
- **Active SIP Accounts:** Growth from ~4 crore to nearly 10 crore accounts.
- **YoY SIP Growth Rate:** Peaked near 60% in 2024 before stabilizing around 20%.
- **SIP AUM Growth:** Expanded from ~₹5 lakh crore in 2022 to ~₹20 lakh crore in 2025.
- **New SIP Account Additions:** Significant spike observed in mid-2025.

Insights

- SIP inflows have shown **consistent growth**, reflecting strong retail investor confidence.
- The number of active SIP accounts nearly **doubled** over the analysis period.
- Growth rates peaked in 2024, indicating a surge in investor participation before stabilizing.
- SIP AUM expansion demonstrates the increasing role of systematic investments in wealth creation.
- New account additions highlight continued penetration into retail markets.

SIP Analysis Dashboard



Dashboard – Category Inflow Analysis

This dashboard presents insights into **mutual fund category inflows**, highlighting how different investment categories contribute to overall industry growth. It enables comparison across categories and tracks net inflow trends over time.

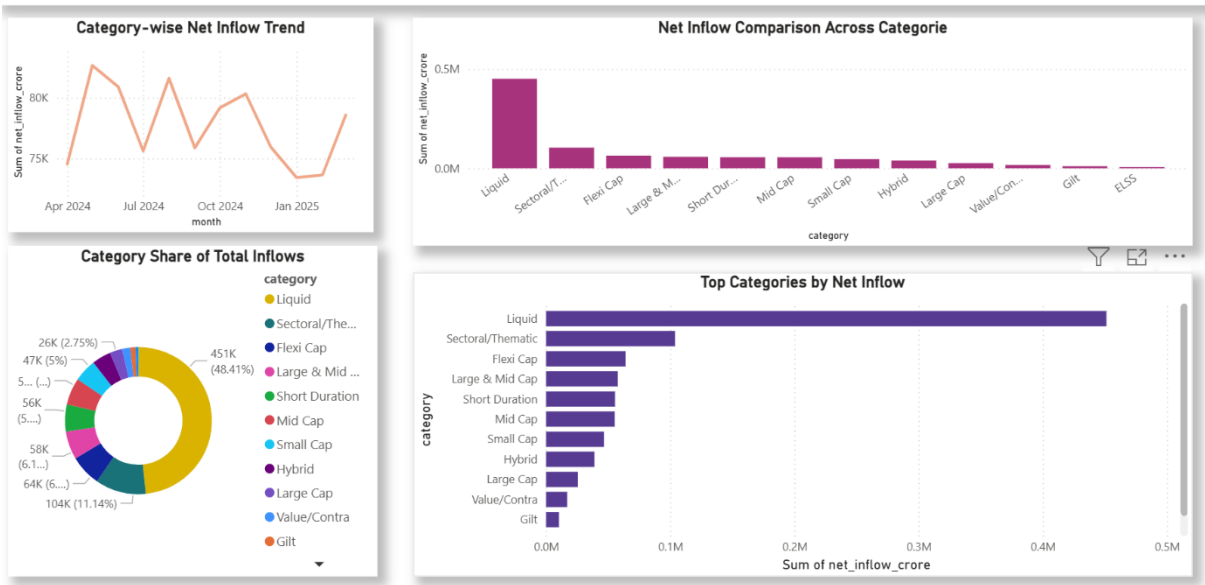
Key Visuals

- **Category-wise Net Inflow Trend:** Line chart showing inflow fluctuations between April 2024 and January 2025.
- **Net Inflow Comparison Across Categories:** Bar chart comparing categories such as Liquid, Flexi Cap, Large & Mid Cap, Mid Cap, and Sectoral/Thematic.
- **Category Share of Total Inflows:** Donut chart showing Liquid funds leading with ~48% share, followed by Flexi Cap and Mid Cap.
- **Top Categories by Net Inflow:** Ranking chart confirming Liquid as the dominant category.

Insights

- **Liquid funds** dominate inflows, accounting for nearly half of the total.
- **Flexi Cap** and **Mid Cap** categories show strong participation, reflecting investor appetite for equity diversification.
- **Sectoral/Thematic funds** contribute meaningfully, indicating targeted investment strategies.
- Overall inflows remain stable, with Liquid funds acting as the primary driver of industry growth.

Category Inflow Analysis Dashboard



Dashboard – Scheme Performance Analysis

This dashboard provides a detailed view of **mutual fund scheme performance**, comparing returns, risk-adjusted metrics, and category distribution. It helps investors evaluate schemes across multiple dimensions such as returns, Sharpe ratio, and alpha.

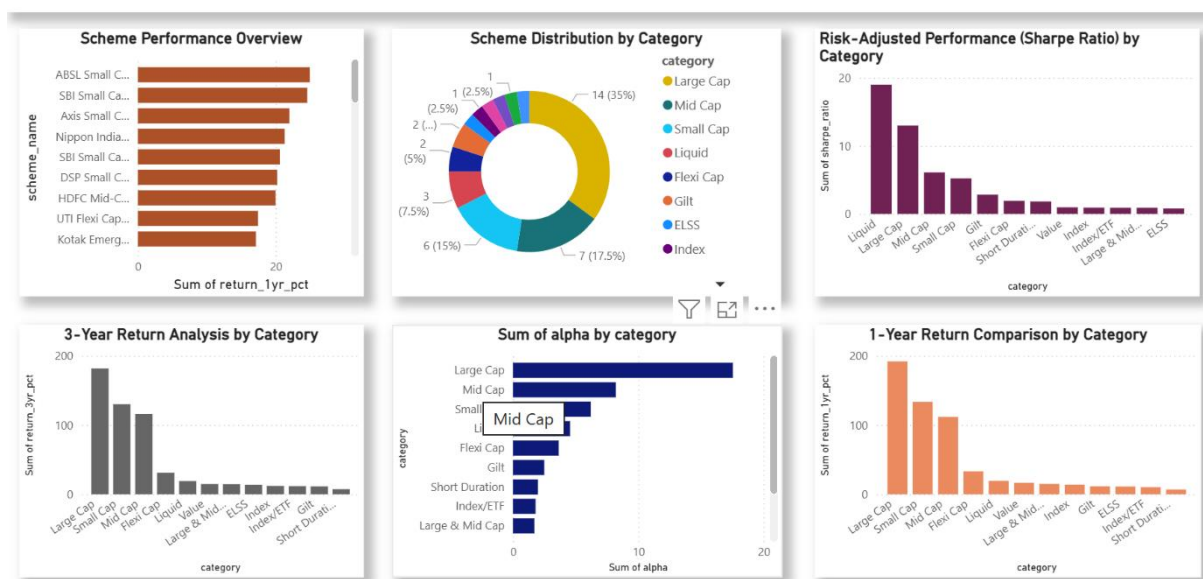
Key Visuals

- **Scheme Performance Overview:** Ranking of schemes by 1-year returns.
- **Category Distribution:** Large Cap dominates (~35%).
- **Sharpe Ratio by Category:** Highlights efficiency of risk-adjusted returns.
- **3-Year Return Analysis:** Long-term performance comparison.
- **Alpha by Category:** Shows benchmark outperformance.
- **1-Year Return Comparison:** Short-term return distribution.

Insights

- **Small Cap schemes** deliver high returns but with higher risk.
- **Large Cap schemes** provide balanced performance and dominate in share.
- **Sharpe ratio** shows Liquid and Large Cap funds as efficient.
- **Alpha values** highlight Mid Cap and Flexi Cap outperforming benchmarks.

Scheme Performance Analysis Dashboard



Key Findings

Major Findings

- Strong growth observed in **SIP investments**, with inflows crossing ₹31,002 crore in Dec 2025.
- **Risk-adjusted metrics** such as Sharpe ratio and alpha help identify quality funds.
- **Investor activity** has increased steadily, with folios expanding to 26.12 crore.
- Dashboards enable **quick performance comparison** across categories, schemes, and fund houses.
- Data visualization improves **decision-making efficiency** for investors and analysts.
- **Small-cap funds** deliver high returns but are highly volatile.
- **Debt funds** provide stability, suitable for risk-averse investors.
- **Balanced funds** offer an optimal trade-off between risk and return.
- **Diversification** reduces portfolio risk and enhances stability.
- **Category inflows** show Liquid funds dominating, while Flexi Cap and Mid Cap attract strong equity participation.
- **Scheme performance analysis** highlights that Mid Cap and Flexi Cap categories outperform benchmarks on alpha values.
- **Market share analysis** confirms SBI Mutual Fund as the industry leader, followed by ICICI Prudential and HDFC.

Conclusion & Future Scope

Conclusion

The **Mutual Fund Analytics Platform** successfully integrates **data engineering, analytics, and visualization techniques** into a unified solution. By leveraging Python for ETL processes, SQLite for structured data storage, and Power BI for interactive dashboards, the project demonstrates how technology can transform raw financial data into actionable insights.

The platform provides meaningful analysis of **fund performance, investor behavior, and market trends**, enabling stakeholders to make informed decisions. Risk-adjusted metrics such as **Sharpe ratio** and **alpha values** help identify quality funds, while dashboards simplify complex data into clear visualizations. Overall, the project showcases the practical application of analytics in the financial domain and highlights the growing importance of data-driven decision-making in mutual fund investments.

Future Enhancements

To further strengthen the platform, several enhancements can be implemented:

- **Real-time data integration** Incorporating live feeds from AMFI, NSE, and BSE will allow dashboards to reflect up-to-the-minute market changes, improving relevance for investors and analysts.
- **Predictive analytics models** Machine learning algorithms can forecast fund performance, investor inflows, and market trends, enabling proactive investment strategies.
- **Portfolio optimization features** Advanced optimization techniques such as mean-variance analysis and risk-parity models can help investors construct portfolios tailored to their risk appetite.
- **Web-based dashboard deployment** Hosting the dashboards on a secure web platform will make them accessible to a wider audience, including investors, fund managers, and researchers, ensuring scalability and usability.